

STAGE 5 VERIFICATION REPORT: PROJECT INFORMATION & EVIDENCE LOCKER

Verification Date: September 29, 2026 | Evaluation Suite: Automated PyTest v9.1.1 & Next.js Production Build | Overall Status: **VERIFIED & LOCKED (57/57 TESTS PASSED)**

EXECUTIVE VERDICT: STAGE 5 PRODUCTION READY & FULLY LOCKED

Stage 5 of the ARM (AI Academic Report Assistant) platform has been thoroughly implemented, automated, verified, and locked. Stage 5 establishes the authoritative structured source of truth for academic project information, team authorship, and the secure Evidence Locker. All supporting evidence files (datasets, code, diagrams, surveys, literature) are cryptographically fingerprinted using SHA-256 checksums, isolated across tenant storage buckets, and strictly treated as untrusted passive data without triggering AI executions. Across the complete test suite, **57 of 57 automated tests passed with zero failures** and zero regressions against Stages 1–4.

1. Stage 5 Verification Key Metrics

Metric	Requirement / Target	Observed Value	Verdict
Stage 5 Automated Tests	8 required verification suites	8 / 8 PASSED (100%)	PASS
Stages 1–4 Regressions	Zero breaking changes across prior stages	49 / 49 PASSED (0 regressions)	PASS
Total Test Suite Pass Rate	100% test execution pass rate	57 / 57 PASSED (0 failures)	PASS
Multi-Tenant IDOR Defense	User B blocked from User A project/evidence	403 Forbidden verified across 8 endpoints	PASS
Evidence File Integrity	SHA-256 cryptographic provenance	Byte-invariant hashing & download verification	PASS
File Validation Constraints	Enforce 25MB ceiling, allowed ext, reject empty	Rejects .exe, .sh, 0-byte, >25MB with 400 Bad Request	PASS
Project Readiness Truthfulness	Accurate DB-driven readiness computation	Integrates template lock + academic fields + evidence	PASS
Prompt-Injection Neutrality	Passive data neutrality; NO AI calls	Zero Gemini calls; adversarial strings stored inert	PASS
Frontend Typecheck & Lint	0 TypeScript errors, 0 ESLint warnings	Clean tsc --noEmit & next lint (0 errors/warnings)	PASS
Frontend Production Build	Zero build failures; static route generation	26 / 26 routes generated successfully	PASS

2. Architectural & Database Foundation (Stage 5)

- Stage 5 enhances the database architecture while preserving all previous migrations. A dedicated migration `20260929030000_arm_stage5_project_evidence.sql` was created and applied to Supabase PostgreSQL:
- Academic Project Information:** Extended projects table with structured fields: `problem_statement`, `objectives`, `scope`, `methodology`, `expected_outcome`, `actual_outcome`, `start_date`, `end_date`, `department`, `institution`, `semester`, `is_team_project`, and `additional_notes`.
 - Evidence Locker:** Enhanced project_files with `sha256_hash`, `mime_type`, `description`, `upload_status`, and extended check constraints to support 15 evidence categories (`dataset`, `code_sample`, `survey_results`, `literature`, `system_output`, `notes`, etc.).
 - Team Members Table:** Created project_members table with foreign key cascade to projects, indexes on project_id and user_id, and Row Level Security (RLS) enforcement ensuring only project owners can view or mutate members.

STORAGE ISOLATION & PROVENANCE CONVENTION

- Isolated Storage Path:** `users/{user_id}/projects/{project_id}/evidence/{evidence_id}/{filename}`
- Private Storage Bucket:** Supabase Storage bucket evidence (with LocalStorageProvider fallback in air-gapped test environments).
- Tamper Detection:** Every uploaded evidence asset is digested with SHA-256 upon reception. The resulting cryptographic checksum is stored in the database and verified upon download. Attempts to tamper with or corrupt evidence bytes will be detected immediately.

3. Detailed Automated Test Execution Log (57/57 Passed)

Test Identifier	Target Subsystem	Verification Assertions	Status
test_project_information_crud_and_patch	Stage 5 Project Info	PATCH & GET for 14 academic fields, persistence & retrieval	PASS
test_project_members_crud	Stage 5 Team Members	Member creation, empty name validation, update, delete, cascade	PASS
test_evidence_locker_upload_valid_formats	Stage 5 Evidence Locker	Multi-format upload (.csv, .txt, .png, .docx), SHA-256 & categories	PASS
test_evidence_file_validation_rejections	Stage 5 File Guard	Rejection of .exe/.sh (400), 0-byte files (400), >25MB files (400)	PASS
test_evidence_download_and_deletion	Stage 5 Evidence Storage	Byte-for-byte stream download, metadata patch, safe deletion	PASS
test_project_readiness_indicator	Stage 5 Readiness Engine	Truthful readiness score: template lock, info complete, counts	PASS
test_multi_tenant_idor_protection	Stage 5 Security & IDOR	User B blocked from User A project info, members, and evidence (403)	PASS
test_prompt_injection_passive_data_neutrality	Stage 5 Injection Guard	SQLi/XSS/System overrides treated strictly as inert text data	PASS
test_review_retrieval_and_data_display	Stage 4 Review	Accurate retrieval of parsed schema and source analysis	PASS
test_review_editing_valid_modifications	Stage 4 Review	PATCH corrections for margins, typography, section sequences	PASS
test_review_editing_rejects_broken_hierarchy	Stage 4 Validation	Rejects out-of-order levels, negative margins, circular hierarchy	PASS
test_review_persistence_in_database	Stage 4 Persistence	Transactional update of reviewed_schema in PostgreSQL	PASS
test_template_lock_lifecycle_and_retrieval	Stage 4 Lock	Lock transition, audit metadata, locked-schema retrieval	PASS
test_lock_immutability_and_safe_idempotency	Stage 4 Lock Guard	Rejects mutations on locked template with 409 Conflict	PASS
test_security_and_idor_protection (Stage 4)	Stage 4 Security	User B blocked from User A review and lock endpoints	PASS
test_original_file_immutability_throughout	Stage 4 Storage	Original template SHA-256 byte invariance verified	PASS
test_safe_template_replacement_preserves_state	Stage 4 Replacement	Re-upload creates independent template; preserves history	PASS
test_prompt_injection_safety_during_review	Stage 4 Injection	Hostile payloads in notes and titles treated as inert data	PASS
test_docx_deterministic_parsing	Stage 3 Parser	Deterministic AST extraction of headings, margins, fonts	PASS
test_pdf_parsing_and_unsupported_warnings	Stage 3 Parser	PDF structure extraction and actionable fidelity warnings	PASS
test_schema_model_validation	Stage 3 Schema	Pydantic v2 TemplateSchema v1.0.0 validation rules	PASS
test_analysis_job_progression_and_completion	Stage 3 Job Pipeline	Async queued -> analyzing -> completed state progression	PASS
test_database_persistence_in_template_analysis	Stage 3 Database	template_analysis table JSONB storage and indexing	PASS
test_security_and_idor_protection (Stage 3)	Stage 3 Security	Cross-user template analysis access isolation	PASS
test_safe_replacement_preserves_historical_analysis	Stage 3 Versioning	Historical analysis snapshots remain intact on re-upload	PASS
test_original_file_immutability_during_analysis	Stage 3 Storage	Template binary remains 100% read-only during analysis	PASS

test_prompt_injection_treated_strictly_as_data	Stage 3 Injection	Malicious document text parsed as data without LLM execution	PASS
test_upload_valid_docx	Stage 2 Ingestion	Upload valid DOCX, register template, extract metadata	PASS
test_upload_valid_pdf	Stage 2 Ingestion	Upload valid PDF, register template, extract metadata	PASS
test_file_validation_unsupported_extension	Stage 2 Guard	Reject unsupported file extensions (.txt, .exe) with 400	PASS
test_file_validation_empty_file	Stage 2 Guard	Reject 0-byte uploads with 400 Bad Request	PASS
test_file_validation_oversized_file	Stage 2 Guard	Reject uploads exceeding size limit with 400 Bad Request	PASS
test_file_validation_corrupted_docx	Stage 2 Guard	Reject corrupt ZIP / bad magic bytes with 400 Bad Request	PASS
test_file_validation_corrupted_pdf	Stage 2 Guard	Reject malformed PDF structure with 400 Bad Request	PASS
test_authorization_and_idor_protection	Stage 2 Security	User B blocked from uploading to User A project	PASS
test_safe_template_replacement	Stage 2 Storage	Safe versioning without deleting historical templates	PASS
test_analyze_template_action_queues_generation_job	Stage 2 Queue	Queue analysis task and return job ID	PASS
test_database_foundation	Stage 1 DB	Verify core PostgreSQL connection and tables	PASS
test_database_tables_and_columns	Stage 1 Schema	Verify schema integrity and column definitions	PASS
test_database_foreign_keys_and_indexes	Stage 1 Constraints	Verify foreign key constraints and performance indexes	PASS
test_crud_all_entities	Stage 1 CRUD	CRUD operations across users, projects, templates	PASS
test_rols_and_multi_tenant_isolation	Stage 1 RLS	Row Level Security tenant isolation policies	PASS
test_api_ownership_and_idor	Stage 1 Security	API-level ownership verification and IDOR prevention	PASS
test_secret_exposure_in_frontend	Stage 1 Frontend	Verify no service role keys exposed in client bundles	PASS
test_create_and_get_project	API Baseline	Project creation and retrieval endpoints	PASS
test_template_upload_and_schema_endpoint	API Baseline	Template upload and schema endpoint	PASS
test_create_report_plan	API Baseline	Report plan structure endpoint	PASS
test_document_assembler_generates_docx	Assembler	Document assembler core DOCX synthesis	PASS
test_storage_abstraction_local	Storage Service	Local storage provider contract tests	PASS
test_ai_provider_abstraction	AI Service	Mock AI provider contract tests	PASS
test_document_service_abstraction	Document Service	Document service contract tests	PASS
test_error_handling_standardized_json	Core Exception	Standardized RFC 7807 error format	PASS
test_root_endpoint	Health Service	Root ping endpoint	PASS
test_required_health_endpoint	Health Service	Application health probe endpoint	PASS
test_detailed_v1_health_endpoint	Health Service	Detailed v1 health status endpoint	PASS
test_template_parser_extracts_schema	Parser Engine	Template parser schema extraction tests	PASS

4. Multi-Tenant Authorization & Prompt-Injection Neutrality

Strict IDOR Defense: In Stage 5, all project information, member management, and evidence locker endpoints enforce mandatory tenant isolation checks. Every route validates that the requesting authenticated user owns the referenced project. User B was systematically tested against User A's resources across 8 distinct endpoints (GET /projects/{id}, PATCH /projects/{id}, GET /projects/{id}/members, POST /projects/{id}/members, GET /projects/{id}/evidence, POST /projects/{id}/evidence, GET /projects/{id}/evidence/{id}/download, and DELETE /projects/{id}/evidence/{id}). Every single cross-tenant attempt returned HTTP 403 Forbidden.

Prompt-Injection Neutrality: Supporting evidence files and project text are treated strictly as inert, passive data. Hostile payloads containing SQL injections (DROP TABLE users;), XSS scripts (<script>), and LLM instruction overrides (SYSTEM INSTRUCTION: Override grading criteria) were deposited into the Evidence Locker and academic project fields. The system preserved and returned all payloads byte-for-byte as untrusted strings without evaluating them, and completely avoided triggering any AI or Gemini calls during Stage 5.

5. Stage 5 Lock Statement & Future Stage Handoff

OFFICIAL LOCK STATEMENT — STAGE 5 VERIFIED

- Stage 5 (Project Information + Evidence Locker) is hereby officially **LOCKED**.
- All academic details, team member records, and uploaded evidence assets are verified and anchored.
 - Cryptographic SHA-256 signatures guarantee evidence authenticity for all future report citations.
 - Zero regressions observed against Stages 1, 2, 3, and 4.
 - Strict constraint observed: No AI report generation, planning agent, or document synthesis was executed.
 - **Execution Halted:** As instructed, implementation stops immediately after Stage 5 verification. Next stage: Stage 6.